

Diego Fernando Lojan Tenesaca

Data & AI Engineer | Full Stack Developer

Applied AI Engineer at Inti Horizons IA

📍 Quito, Ecuador | 📞 +593 939 327 368 | ✉ fernando.lojan10@gmail.com

🌐 diegofernandolojantenesaca.github.io | 🏠 github.com/DiegoFernandoLojanTenesaca | 🔗 linkedin.com/in/diego-fernando-lojan

🔧 gitlab.com/fernando.lojan10 | 🏆 kaggle.com/diegofernandoljtn | 🆔 0009-0003-3882-3889

Professional Summary

Computer Science Engineer with four years developing software, the last two professionally, and two focused on machine learning, data science and natural language processing (NLP). I was part of the team behind the AI microservice of a live product, with multi-tenant architecture, 18 functional domains, a 15-tool agent and semantic search over 139,000 vectorized conversations on pgvector. My work has been taking models to production and keeping them there. I build ETL and streaming data pipelines, work with RAG and hybrid retrieval, orchestrate LLMs and vision-language models with LangGraph, and run accelerated inference on GPU and on-device with ONNX Runtime, down to quantizing a classifier to 3.8 MB so it runs inside the browser with no connection. I also develop the web and mobile applications that consume all of it. I have a paper accepted at Springer Nature (CIT 2026) on stress detection in Spanish.

Publications

- **Lojan, D. F.; Chamba-Eras, L.** (2026). *Bayesian Hyperparameter Optimization of Support Vector Machines Using TPE for Stress Detection in Spanish Social Media During the 2024 Ecuadorian Energy Crisis*. In *Emerging Research in Intelligent Systems*, proceedings of the Congress on Innovation and Technology (CIT 2026), *Lecture Notes in Networks and Systems* series, Springer Nature. (Accepted, in production)

Professional Experience

Inti Horizons IA | *Applied AI Engineer*
Cuenca, Ecuador / Remote

Oct 2026 - Present

Indaga Lab | *Founder & Lead Developer*
Quito, Ecuador / Remote

2025 - Present
Own software and AI product studio

- **Wauto Indaga:** AI-powered CRM and messaging automation for WhatsApp, Telegram and Meta, live in production with five commercial tiers and agent replies under 2 seconds. Hybrid retrieval (full-text + pgvector), model cascade with failover, versioned public API and MCP server.
- **Kumbre:** converts construction bids from PDF (Ecuadorian public procurement format) into editable Excel through intelligent parsing and fuzzy matching, with a preloaded catalog and official price indexes.
- **AgentOS:** Android app that turns Google-free phones into a 24/7 AI agent operated over Telegram. Released as open source.
- Technical leadership of the team, product definition and pricing model.

Xynitra Devs | *Co-founder*
Quito, Ecuador

Jul 2026 - Present
Software development studio

- OTP code platform for streaming accounts: email intake through Cloudflare Email Workers, real-time code extraction, Upstash Redis as the store, a spreadsheet as the source of truth for accounts, CAPTCHA and an admin panel with a killswitch. Next.js 16 and FastAPI.
- Technical direction of the studio and its products alongside the founding team.

Freelance Developer | *Software & AI Solutions*
Remote

2024 - Present

- Conversational bots (Telegram, WhatsApp) with LLM orchestration, cross-provider failover, versioned prompts and strict output validation. Lead qualification, CRM and customer support flows.
- CRM for an energy advisory firm, plus shared-account management platforms with automatic OTP extraction, email webhooks and an admin panel.
- Anomaly detection over multi-country time series, data feasibility studies with web scraping, and data automation with n8n.
- Computational support for theses and academic research: numerical simulation of seismic isolators for a Civil Engineering master's thesis, with equivalent Python and MATLAB implementations validated against each other, plus the figure and table package for the committee.

- AI microservice built on DDD with schema-isolated multi-tenancy in PostgreSQL, organized into **18 functional domains** behind a single REST API.
- Semantic search over **139,000 vectorized conversations** using multilingual BGE-M3 embeddings (1,024 dimensions) and an HNSW index in pgvector.
- ETL pipelines with async jobs and background workers. Re-embedded **62,000 conversations in under an hour** on an NVIDIA A6000 GPU with Text Embeddings Inference.
- In-house benchmark of vector engines across the full 139,000 real vectors (latency, recall and memory) to pick the production engine.
- Conversational agent with **15 tool-calling tools**, loop control, normalized caching and PII masking, orchestrated with LangChain and LangGraph.
- Self-hosted Vision-Language Model pipelines on GPU (Qwen-VL, InternVL) with batch processing, checkpoints and retries. Zero-shot classification, OCR, local audio transcription and offline translation services.
- Reversible migrations with checksums, model observability (token and latency tracing), per-tenant rate limiting and containerized deployment with CI/CD.

- Next.js web back office and cross-platform React Native mobile app, with dynamic theming and internationalization.
- Domain-partitioned Python REST API with JWT + OAuth authentication and a granular role and permission model.
- PostgreSQL query optimization, real-time messaging over WebSockets and Redis caching.
- Third-party integrations (WhatsApp Business API, Google Calendar, payment gateways, OTP codes), a visual conversational flow builder and OCR document processing.

Featured Projects

- **Sudial AI**: Production AI microservice: domain-partitioned API, an embedding base over real conversations, a tool-orchestrating agent and observability. FastAPI, LangGraph, pgvector.
- **Riksi**: Identifies Ecuadorian wildlife and plants inside the browser with no connection, using a 3.8 MB quantized model over 100 species. Rounded out by **Yachaq**, a seven-tool agent with RAG and a provider cascade that lives in 512 MB of RAM, and **riksi-radar**, a Kafka → DuckDB → dbt pipeline that pits the model against GBIF's human labels and confirms it holds up out of distribution. ONNX Runtime, FastAPI, MCP.
- **Wauto Indaga**: AI-powered CRM and messaging automation for WhatsApp, Telegram and Meta: multi-agent shared inbox, Kanban pipelines, visual automation builder, hybrid knowledge base and MCP server. Next.js 16, React 19, Supabase.
- **CConnect** (open source, contributor): Mobile, desktop and web client for driving Claude Code on another machine: chats in tabs, interactive permissions as forms, a file manager, a monitor and a terminal. FastAPI backend over the Agent SDK, published through a Tailscale Funnel or behind your own proxy. I contributed to the backend and the app, first in Compose Multiplatform and then through its migration to Svelte + Tauri.
- **Ordo**: desktop app that organizes Gmail with AI. It proposes a label per sender and turns it into Gmail filters that keep working with the computer off. Runs on Claude, Groq, local Ollama or any OpenAI-compatible API; the AI only proposes, its output is validated and the user approves every change. Rust, Tauri, Svelte.
- **Xyra** (creator, with Xynitra Devs): Desktop and Android companion for League of Legends: connects to the local client over WebSocket, reads augment cards with OCR and pairs the phone by QR code. SHA-256-verified in-app updates and tag-driven CI releases; under review on the Riot Games developer portal. Rust, Tauri 2, Svelte 5, Kotlin.
- **Indaga Lab**: The brand's own platform: service catalog, a quoting engine with a four-provider LLM cascade, and internationalization. Next.js 16, FastAPI, Neon Postgres.
- **LATAM Energy Anomalies**: Energy crisis detection across 8 countries through consensus of three techniques (Isolation Forest, STL, CUSUM), benchmarked against nine baselines and validated with bootstrap CIs and McNemar's test. F1 of 0.75. Scikit-learn, Optuna, SHAP, Streamlit.
- **Stress Detection in Spanish**: Code and experiments behind the paper accepted at CIT 2026: SVM tuned with Optuna TPE under the CRISP-DM methodology, over posts from Ecuador's 2024 energy crisis.
- **Cuadrekit**: Free toolkit for Ecuadorian micro-businesses: severance calculations, quotes, receipts and payroll, offline and subscription-free. Vue.

More projects, including AgentOS, Yapa, Reps, ADBox and OfficeGo, at github.com/DiegoFernandoLojanTenesaca and in the [portfolio](#).

Version Control & Change Management

- **Git**: daily work on monorepo and multi-module repositories. Feature branches, rebasing and clean integration, conflict resolution, conventional commits, version tags and a *CHANGELOG* per release. History as documentation: every change traceable to its reason.
- **GitHub and GitLab**: Pull Requests and Merge Requests with code review, **GitHub Actions** and CI/CD pipelines into containerised deployment, versioned Releases and GitHub Pages. Including contributions to an open source project reviewed by others.
- **Change management**: a version contract across modules (server, app and CLI) with declared compatibility and out-of-date warnings for the user, deployments with rollback, and technical ownership of my own team’s workflow.

Education

Computer Science Engineer – Universidad Nacional de Loja	Loja, Ecuador
International Studies in Systems & Computer Engineering (1 term) – UTP	Lima, Peru
Diploma in Project-Based Learning (120 h) – Politécnico de Colombia	Online, 2026
Fifth School of Quantum Computing (28 h) – UNAM, School of Engineering	Mexico City, 2026

Technical Skills

AI / ML: LLMs (LangChain, LangGraph) | RAG and hybrid retrieval | Tool-calling agents | MCP | Embeddings and Vector Search | NLP | Zero-shot classification | Anomaly Detection | OCR | Speech-to-Text | Vision-Language Models | ONNX Runtime | int8 quantization and on-device ML

Data: SQL | ETL | Kafka | DuckDB | dbt | MLflow | EDA and Data Wrangling | Time Series | Web Scraping | Statistical Analysis | Pandas | NumPy | SciPy | Streamlit | Plotly | MATLAB

Backend & Databases: Python (FastAPI, Pyramid, Django) | Rust | Node.js | Java (Spring Boot) | REST APIs | Microservices | WebSockets | OAuth 2.0 / JWT | PostgreSQL (+pgvector) | Redis | Supabase | SQLAlchemy

Frontend & Mobile: React | Next.js | TypeScript | Svelte | Tailwind CSS | Astro | Nuxt / Vue | PWA | Kotlin | Jetpack Compose | Compose Multiplatform | React Native / Expo | Tauri (Rust)

Version Control: Git (branching, rebase, conflict resolution, tags and releases) | GitHub (Actions, Pull Requests, Releases, Pages) | GitLab (CI/CD, Merge Requests) | Code review | Semantic versioning and CHANGELOG

DevOps & Tooling: Docker | Docker Compose | CI/CD | Linux (CLI) | Server and VPS administration | Nginx / Caddy | Claude Code | Agent SDK | MCP servers | n8n

Networking & Security: Cybersecurity fundamentals | Network defense | Endpoint security | Threat management | IPv6 | OAuth 2.0 / JWT | PII masking | Backup and recovery

Certifications

Python for Data Science (LinkedIn Learning, 2026): [Advanced Data Scientist](#), [Big Data Essentials](#), [PySpark](#), [NLTK](#).

Generative AI & Agents (LinkedIn Learning, 2026): [AI Agent Foundations \(Architectures, Frameworks and Applications\)](#), [Generative AI Foundations](#), [What Is Generative AI](#), [Programming Foundations: AI](#).

Data Analysis (Microsoft & LinkedIn, 2026): [Career Essentials in Data Analysis](#) (5-course path), [Microsoft Fabric](#).

Networking & Cybersecurity (Cisco Networking Academy, via Universidad Tecnológica del Perú, 2025): [Introduction to Cybersecurity](#), [Endpoint Security](#), [Network Defense](#), [Cyber Threat Management](#). [IPv6](#) (Sage, 2023).

Languages

Spanish: Native | English: Intermediate (B1), fluent technical reading and written communication ([view certificate](#))